

Sean Aidan O'Toole

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EDUCATION

IE University

Master in Business Analytics and Data Science

Madrid, Spain

Sept. 2026 – Jul. 2027

University of Texas at San Antonio

Bachelor of Science in Computer Science, Minor in Mathematics

Concentration in Data Science and Software Engineering

San Antonio, TX

Aug. 2022 – May 2026

GPA: 3.90

SKILLS

Languages: Java, Python, C, SQL, JavaScript, PHP, HTML/CSS, R, BASIC, BASH, Kotlin

Developer Tools: Cloud Technologies (AWS), Git, Android Studio — **Libraries:** TensorFlow, SKlearn, Matplotlib

Other: German (Intermediate), Spanish (Elementary), Adobe Photoshop and Premiere Pro (2000+ hours)

EXPERIENCE

Database Developer

The Experts Tribe

May 2025 – August 2025

San Antonio, TX

- Designed and maintained relational and cloud-hosted databases (PostgreSQL, MySQL, Firebase Firestore).
- Implemented efficient schema structures and data models to ensure seamless integration with frontend components and scalability for future feature expansion.
- Optimized queries and data retrieval processes, reducing potential latency and improving overall user experience.

Artificial Intelligence Research Assistant

University of Texas at San Antonio

Oct 2024 – May 2025

San Antonio, TX

- Utilized Llama to create automated feedback loops for software development tasks, enhancing workflow efficiency.
- Analyzed and categorized GitHub change requests, identifying patterns that led to a 60% increase in model accuracy for understanding and improving developer feedback.
- Developed scripts to extract user comments and change requests from GitHub repositories via BeautifulSoup

PROJECTS

UTSA GPT | *Python, FastAPI, RAG, Gemini, Firestore*

- Owned the Python backend for a RAG chatbot grounding Gemini on private Firestore data via async FastAPI.
- Fused dense vector embeddings with BM25 keyword scoring, reranking the top 10 through a cross-encoder to 5.
- Cached embeddings by SHA-256 hash and moved four blocking calls to threads, removing event-loop stalls.
- Fixed a tokenizer defect that silently disabled sparse retrieval, lifting ranked coverage from 6% to 100%.

High-Volume Data Ingestion & Modeling Pipeline | *AWS, PHP, MySQL*

- Designed and implemented a scalable ETL pipeline to process and validate 5M+ records using parallel file chunking and multi-process execution.
- Developed a normalized relational schema by decomposing device and manufacturer attributes into indexed lookup tables, reducing storage overhead by 90% and improving upload speed by 400%.
- Implemented batch-based insertion with SQL constraint handling, enabling efficient duplicate detection and automatic routing of invalid records to a structured error table.

Predictive Analytics Project | *Python, scikit-learn, React, FastAPI*

- Built a fantasy draft assistant with dynamic pick recommendations, draft grades, and post-draft analysis.
- Created an ETL pipeline collecting Sleeper ADP, historical player data, seasonal projections, and league scoring settings into a SQLite relational database.
- Deployed a full-stack app with a React frontend and FastAPI backend serving data and recommendations.

AWARDS AND HONORS

Positions: ACM Media Officer (2023-2025), RowdyHacks Officer (2023-2025), GWG Scholar (2022-2025)

Certifications: Google Data Analytics Professional, IBM AI Foundations for Business, U. Michigan Generative AI

Honors: President's Scholarship (5x), Dr. Craig Endowed Scholarship in the Sciences, Most Outstanding IT Student